

Enhancing Four-Class Motor Imagery Detection through Advanced Feature Extraction Techniques

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Abstract—Brain-computer interface (BCI) technology has great potential in control, communication, and neurological diagnostics by interpreting EEG signals. This paper introduces a novel method for extracting distinguishing features from EEG signals using a Common Spatial Patterns (CSP) model, enhanced with a Phase Space Reconstruction model and followed by an autoregressive (AR) model. Initially, EEG signals are denoised using an IIR Butterworth filter before feature extraction. Two classifiers, Linear Discriminant Analysis (LDA) and Multi-Layer Perceptron (MLP), are employed to differentiate features across four motor imagery (MI) tasks. For a four-class MI task, combining spatial and temporal-frequency domain features, LDA achieves an accuracy of 67.40%, while MLP reaches 64.78%. LDA yields precision and recall scores of 67.40% and 67.20%, respectively, for the F1-score, whereas MLP records precision, recall, and F1-scores of 65.36%, 64.78%, and 64.28%, respectively. The proposed architecture is evaluated using the BCI Competition IV set 2a dataset, proving its effectiveness in EEG signal classification for BCI applications.

Index Terms—Brain-Computer Interface ;EEG ; Motor Imagery; Feature extraction; Multi-classification

I. INTRODUCTION

Brain-Computer Interfaces (BCI) represent a revolutionary technology that translates human brain activity into control commands [2, 13] and also in automatic diagnosis [12, 20], establishing a direct link between the mind and external devices. This innovation significantly enhances communication by allowing users to control various systems without relying on physical movements, bypassing the need for muscle engagement, such as using arms or feet [6, 16]. Motor imagery (MI), which involves controlling limbs or devices through imagined movements, is central to this process, with EEG being the most widely used method for BCI applications.

The literature highlights significant progress in multi-class BCIs, particularly in MI tasks. Various signal processing techniques, such as filtering, feature extraction, and classification, are employed to process MI-EEG signals. Pre-processing typically involves FIR or IIR bandpass filters for noise reduction[3, 9]. Feature extraction spans temporal,

frequency, tempo-frequential, and spatial domains[18, 19]. Classification tasks leverage both deep learning models like Convolutional Neural Networks (CNNs) and classical machine learning techniques such as Support Vector Machines (SVM) and Linear Discriminant Analysis (LDA) [15]. Noteworthy advancements include the creation of 2D images for CNN classification and the use of wavelet decomposition, as for other image processing applications [14], combined with AR modeling for feature extraction [7]. Additionally, fuzzy logic systems are explored for EEG signal classification [17]. These studies collectively contribute to the enhancement of EEG signal processing for MI detection, paving the way for more sophisticated and efficient BCI systems.

A. Motivation

While significant research has been conducted in the field of Brain-Computer Interfaces (BCIs) and motor imagery (MI) detection, much of it has concentrated on binary or two-class classification tasks. However, real-world applications often demand more complex classification capabilities, especially when multiple motor tasks need to be distinguished simultaneously. Tackling the four-class classification problem offers a pathway to broaden BCI research, leading to more versatile and practical solutions that align more closely with real-world needs. By exploring this relatively underexplored area, we aim to provide valuable insights and advancements in BCI technology, ultimately improving its usability and effectiveness across various applications.

Although deep learning models are widely recognized for their performance across numerous tasks, there are situations where machine learning models are more suitable. In our work, particularly when considering physical implementations, the emphasis is placed on the feature extraction phase. The objective is to create a compact characteristic vector from raw data, reducing dimensionality while preserving crucial information. This approach not only enhances the efficiency of the models but also makes them more practical for real-world BCI applications.

B. Contribution

EEG signal processing forms the backbone of many BCI applications, facilitating the analysis of brain activity. To harness brain data effectively, it's crucial to select the appropriate processing methods. The key contributions of this research are:

- Emphasis on enhancing feature extraction by combining spatial features (via Common Spatial Patterns - CSP) and tempo-frequential features (using Phase Space Reconstruction paired with the AR model).
- Rigorous validation of the proposed architecture using two distinct classification algorithms, ensuring robust performance across various scenarios.

These improvements aim to advance the precision and efficiency of BCI systems.

C. Paper organization

The rest of the paper is organized as follows: The proposed framework is presented in Section II. The results are discussed in Sections III, and finally, the paper is concluded in Section IV.

II. METHODOLOGY

Figure 1 illustrates the various stages of the proposed architecture, which include preprocessing, feature extraction, and classification. Initially, EEG data undergoes filtering with an Infinite Impulse Response (IIR) bandpass filter using the Butterworth coefficient [1]. The discriminative components of the EEG signal are then extracted using a Common Spatial Patterns (CSP) model. To further refine feature extraction, the reconstructed phase space is modeled using an Autoregressive (AR) approach. The extracted CSP features, along with coefficients from the phase space and AR model, are used to train Linear Discriminant Analysis (LDA) and Multi-Layer Perceptron (MLP) classifiers. These classifiers are evaluated through 5-fold cross-validation to mitigate overfitting and ensure balanced class distribution. Finally, a voting mechanism is employed to combine the predictions, with the final class determined by majority vote.

A. Butterworth filter

To eliminate artifacts and unwanted noise, EEG signal filtering is applied using a Butterworth filter, chosen for its simplicity and nearly flat amplitude response in the passband. This characteristic is essential for maintaining the integrity of the EEG data. The filter order is set to six, allowing the passage of frequencies between 8 and 30 Hz, which correspond to the sensori-motor rhythm. This range is crucial for capturing relevant brain activity without distorting the signal.

B. Feature extraction

1) *Common Spatial Patterns (CSP)*: CSP is a widely used feature extraction method in motor imagery-based BCIs (MI-BCIs) [5]. It works by learning spatial filters that reduce the variance of one class while increasing the variance of another. Although CSP is traditionally applied to two-class data, variants like the "One-Versus-the-Rest (OVR)" technique

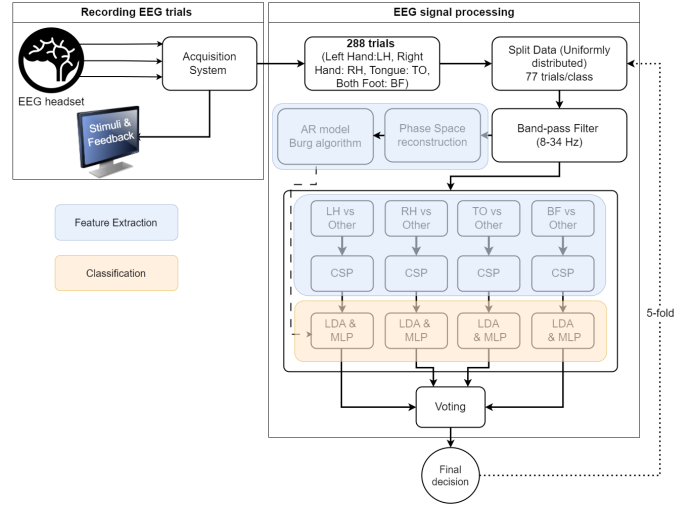


Fig. 1: Proposed architecture for MI-BCI multi-class classification.

extend its use to multi-class data. In multi-class classification, CSP with OVR decomposes the problem into binary sub-problems, extracting distinct spatial filters for each class. Figure 1 illustrates the division of filtered EEG signals into four reactions using the OVR approach, with Algorithm 1 detailing the CSP process repeated for each reaction.

Algorithm 1 Common Spatial Pattern (CSP) Algorithm for EEG Processing

- 1: **Input:** EEG data with two classes ($\mathbf{X}_1, \mathbf{X}_2$)
- 2: **Output:** Projection matrix ($\mathbf{W}$)
- 3: Concatenate trials from both classes: $\mathbf{X} = [\mathbf{X}_1, \mathbf{X}_2]$
- 4: Calculate the covariance matrices for each class:
- 5: $\mathbf{C}_1 = \frac{1}{n_1} \mathbf{X}_1 \mathbf{X}_1^T$
- 6: $\mathbf{C}_2 = \frac{1}{n_2} \mathbf{X}_2 \mathbf{X}_2^T$
- 7: Average covariance matrices:
- 8: $\mathbf{C} = \frac{1}{2} (\mathbf{C}_1 + \mathbf{C}_2)$
- 9: Perform eigenvalue decomposition: $\mathbf{C} = \mathbf{U} \mathbf{\Lambda} \mathbf{U}^T$
- 10: Select the top k and bottom k eigenvectors:
- 11: $\mathbf{U}_k = [\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k, \mathbf{u}_{n-k+1}, \dots, \mathbf{u}_n]$
- 12: Construct the whitening matrix:
- 13: $\mathbf{W}_w = \mathbf{U}_k^T \mathbf{\Lambda}^{-1/2} \mathbf{U}_k$
- 14: Apply whitening to covariance matrices:
- 15: $\tilde{\mathbf{C}}_1 = \mathbf{W}_w^T \mathbf{C}_1 \mathbf{W}_w$
- 16: $\tilde{\mathbf{C}}_2 = \mathbf{W}_w^T \mathbf{C}_2 \mathbf{W}_w$
- 17: Perform eigenvalue decomposition on the combined covariance matrix:
- 18: $\tilde{\mathbf{C}} = \frac{1}{2} (\tilde{\mathbf{C}}_1 + \tilde{\mathbf{C}}_2)$
- 19: $\tilde{\mathbf{C}} = \mathbf{V} \mathbf{\Lambda}_d \mathbf{V}^T$
- 20: Compute the projection matrix: $\mathbf{W} = \mathbf{W}_w \mathbf{V}$

2) *Phase-Space Reconstruction*: The Fast Fourier Transform (FFT) is instrumental in extracting both phase and amplitude information from EEG signals. Despite the challenge of leakage, FFT helps compute the frequency spectrum for these

non-periodic signals [7]. To minimize leakage, we employ a Hamming window and use a segment length of 64 samples at a 250 Hz sampling rate. This method disperses noise evenly and enhances the signal-to-noise ratio (SNR). The extracted phase and amplitude data are then integrated into the EEG signal processing pipeline, as outlined in Method 2. These features are averaged for each channel and combined with Autoregressive (AR) features into a single vector for further analysis.

Algorithm 2 FFT for EEG Signal Processing

- 1: **Input:** EEG signal samples ($x(t)$), Sampling rate (f_s), Window function (Hamming)
 - 2: **Output:** Amplitude ($\alpha(t, f)$), Phase ($\phi(t, f)$)
 - 3: **for** each channel i in EEG data **do**
 - 4: Apply Hamming window to $x(t)$
 - 5: Compute FFT: $X(f) = \text{FFT}(x(t) \times \text{Hamming})$
 - 6: Extract amplitude: $\alpha(t, f) = |X(f)|$
 - 7: Extract phase: $\phi(t, f) = \tan^{-1} \left(\frac{\text{Im}(X(f))}{\text{Re}(X(f))} \right)$
 - 8: **end for**
 - 9: **return** Average amplitude and phase for each channel
-

3) *Autoregressive Model (AR)*: The autoregressive model assumes the EEG signal as a linear combination of the signals at past time points, which provides an efficient representation of EEG signals [7]. This algorithmic block represents the autoregressive model for EEG signals, where the EEG signal ($y(t)$) is modeled as a linear combination of its past value up to order p , with α_i representing the AR coefficients.

Algorithm 3 Autoregressive Model for EEG Signal

- 1: **Input:** EEG signal ($y(t)$), Order of the model (p)
 - 2: **Output:** AR coefficients (α_i)
 - 3: **for** $i = 1$ to p **do**
 - 4: Estimate AR coefficient α_i , where, $y(t) = \sum_{i=1}^p \alpha_i \cdot y(t-i) + \varepsilon_t$
 - 5: **end for**
 - 6: **return** AR coefficients (α_i)
-

4) *PSR-AR Cascade: Enhanced Feature Extraction*: EEG signals for BCI are often the result of complex and potentially chaotic dynamic processes, and their raw representation does not capture these underlying dynamics. The Phase Space Reconstruction (PSR) method reveals the hidden structure of an EEG signal by capturing regularities that are not visible in the raw analysis. By transforming a time series into a multi-dimensional space, PSR uses several time-shifted copies of the same signal to represent the underlying non-linear dynamics. Each dimension of this reconstructed space corresponds to a past state of the signal, shifted in time by a specific parameter called delay, enabling a better understanding of the EEG signal nature.

By applying the AutoRegressive (AR) model to PSR data, the temporal dependency of the reconstructed data is directly modeled. This helps capture fine-grained information and

enriches the feature representation. The AR model can extract useful features such as dominant frequencies and recurrent patterns. The Burg algorithm (referred to as Algorithm 4) is used for implementing the AR model. This algorithm minimizes direct and inverse prediction errors, resulting in a more accurate estimate of AR coefficients. Implementing the AR model with Burg's algorithm ensures a more reliable and robust representation of the data features extracted from the reconstructed space.

Algorithm 4 Autoregressive Model Estimation Using Burg's Algorithm

- 1: **Input:** EEG signal ($x(t)$), Model order (p)
- 2: **Output:** AR coefficients (a_i)
- 3: Initialize AR coefficients $a_i^{(0)} = 0$ for $i = 1$ to p
- 4: Initialize residuals $e_n^{(0)} = x(n)$ for $n = 1$ to N
- 5: **for** $m = 1$ to p **do**
- 6: Compute residual autocorrelations r_k for $k = 1$ to m
- 7: Compute reflection coefficients ρ_m :

$$\rho_m = \frac{r_m - \sum_{k=1}^{m-1} a_k^{(m-1)} \cdot r_{m-k}}{1 - \sum_{k=1}^{m-1} a_k^{(m-1)} \cdot r_{m-k}}$$

- 8: Update AR coefficients $a_k^{(m)}$:

$$a_k^{(m)} = a_k^{(m-1)} - \rho_m \cdot a_{m-k}^{(m-1)}$$

- 9: Update residuals $e_n^{(m)}$:

$$e_n^{(m)} = e_n^{(m-1)} - \rho_m \cdot e_{n-m}$$

- 10: **end for**

- 11: **return** AR coefficients (a_i)
-

The CSP features and average Phase-space are combined into a single feature vector and injected into the classification block, which is based on LDA and MLP.

C. Classification

In our classification approach, we employ Linear Discriminant Analysis (LDA) and Multi-Layer Perceptron (MLP) as the primary classifiers.

1) *LDA*: Linear discriminant analysis is commonly used in brain computer interfaces(BCI) to classify brain signals. It is used to identify a linear combination of features that maximizes the separation between distinct types of signals, which is required for reliable data detection and classification.

2) *MLP*: The multilayer perceptron neural network is a neural network architecture consisting of several interconnected layers of neurons [4, 8]. This structure is widely used due to its ability to model complex relationships between data [10].

III. RESULTS AND DISCUSSION

A. Dataset

The suggested processing chain was assessed and tested using the BCI Competition IV set 2a dataset. This dataset contains information from nine participants who were instructed

to imagine tongue movements, both feet, left hand, and right hands. EEG data was recorded using twenty-two conventional electrodes and three electrooculography electrodes (designated as "EOG-left", "EOG-central," and "EOG-right"). The dataset is divided into training and test sets to evaluate the effectiveness of the classification algorithms. The recordings have a sample rate of 250 Hz.

B. Performance Criteria

The EEG signal processing chain used a bandpass Butterworth filter for pre-processing and incorporated CSP, Phase Space Reconstruction, and an AR model of order 5 for feature extraction. We standardized the extracted data and trained classification models using Linear Discriminant Analysis and Multi-Layer Perceptrons. Performance was evaluated using confusion matrices, accuracy, precision, recall, and F-score[20]:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (2)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (3)$$

$$\text{F1 score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4)$$

C. Performance Evaluation

Integrating Phase Space Reconstruction with AR modeling improved the performance of both classifiers. For the LDA classifier, accuracy, precision, recall, and F1-score showed significant improvements. Specifically, the accuracy increased from 65.28% to 67.40% as depicted in Table II. Similarly, for the MLP classifier, there were notable improvements in accuracy and F1-score, with accuracy rising from 62.31% to 64.78%. These results highlight the effectiveness of combining Phase Space Reconstruction with AR modeling in enhancing the classification performance of EEG data. The LDA model consistently outperformed the MLP model, demonstrating its robustness in this context.

TABLE I: Classification Results Before and After Including Phase Space Reconstruction and AR

Algorithm	Metric	Results of CSP	Results of CSP with AR
LDA	Accuracy	65.28%	67.40%
	Precision	65.28%	67.40%
	Recall	65.28%	67.40%
	F1-score	64.93%	67.20%
MLP	Accuracy	62.31%	64.78%
	Precision	65.74%	65.36%
	Recall	62.31%	64.78%
	F1-score	61.70%	64.28%

The classification results for LDA and MLP, before and after integrating Phase Space Reconstruction with AR modeling, are shown in the table table I. For LDA, the accuracy increased

TABLE II: Comparison of Different Methods

Reference	EEG signal processing chain	Accuracy
[23]	3D CNN	75.01%
[15]	CSP+CWT+CNN	71.25%
[21]	EEGNet	62.56%
[22]	CRAM	59.10 %
[19]	CSD-CSP	58.14%
[11]	MCSP	58.07%
Proposed framework	CSP+Phase Space Reconstruction+AR	67.40%

from 65.28% to 67.40%, while for MLP, the accuracy improved from 62.31% to 64.78%. The LDA model consistently outperformed the MLP model.

Table III presents detailed classification metrics for each subject, illustrating significant variability in performance. For LDA, person 1 achieved a precision, recall, and F1-score of 78.12%, whereas person 9 achieved the highest performance with 89.58%. Subject 2, 5, and 6 had relatively lower accuracies around 40%. On average, LDA achieved an overall accuracy of 67.40%.

These results underscore the variability in performance across subjects, highlighting the importance of considering individual differences when evaluating EEG or BCI models.

Table II compares the proposed framework's classification accuracies to existing cutting-edge approaches. Compared to other methods, this approach demonstrates intermediate performance. For instance, 3D CNN and CSP+CWT+CNN outperform the "Proposed framework" with accuracies of 75.01% and 71.25%, respectively, owing to their sophisticated convolutional neural network architectures that more effectively capture the complex features of EEG signals. Although the overall accuracy is lower than CNN-based approaches, it offers the advantage of providing an important and compact feature vector. However, the Proposed framework shows better performance than techniques like FBCSP, CRAM, WPD + CSP + ANN1, CSD-CSP ,and MCSP, which achieve accuracies ranging from 58.07% to 62.56%. This indicates that integrating phase space reconstruction and AR modeling provides a notable advantage over these traditional methods.

IV. CONCLUSION

This paper presents a novel architecture for EEG signal processing in the context of a BCI with four classes. We comprehensively describe the EEG signal by integrating diverse feature extraction methods and combining features from different domains. Our results demonstrate the efficacy of the LDA classifier, achieving an average accuracy of 67.40%; whereas the MLP classifier yielded accuracy values of 62.81% after. Notably, the accuracy of the LDA classifier varied significantly among subjects, with a low of 30.56% for person two and a high of 83.68% for person 9. These findings emphasize the necessity of considering individual differences in neuronal responses and brain activity patterns when designing and interpreting EEG classification models.

TABLE III: Performance Evaluation of Classification Models

Person	LDA				MLP			
	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
1	78.12	78.12	78.12	77.87	74.31	75.15	74.31	73.51
2	39.93	39.93	39.93	40.05	30.56	31.99	30.56	28.33
3	88.89	88.89	88.89	89.00	82.64	82.85	82.64	82.72
4	61.46	61.46	61.46	61.66	61.46	61.93	61.46	61.55
5	44.79	44.79	44.79	43.39	44.44	43.16	44.44	43.07
6	43.06	43.06	43.06	42.21	46.53	47.07	46.53	45.92
7	75.35	75.35	75.35	75.60	77.08	78.23	77.08	77.12
8	85.42	85.42	85.42	85.36	82.29	83.40	82.29	82.42
9	89.58	89.58	89.58	89.64	83.68	84.43	83.68	83.86
Mean	67.40	67.40	67.40	67.20	64.78	65.36	64.78	64.28

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